Why Linear Regression Minimizes Squared Error

A Probabilistic Explanation

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1 The First Model Everyone Learns

Almost everyone who begins studying AI or Machine Learning encounters Linear Regression as their first serious algorithm.

We are given:

- A training set of examples
- A hypothesis
- And a cost function

Linear Regression Cost Function

This is the familiar least-squares cost function. It models the linear relationship between the input features and the target variable by choosing parameters that minimize the average sum of squared errors. In other words, we aim to fit a straight line through the data cloud such that the total squared deviation between predictions and actual values is as small as possible.

But rarely do we stop and ask:

- Why squared error?
- Why not absolute error?
- Why not power of 4?
- Why not something else entirely?

To answer that, we need to move beyond algebra and enter probability theory.

2 The Hidden Assumption Behind Linear Regression

Linear regression is not just a geometric procedure of drawing the "best fit" line. It is built on a probabilistic assumption about how data is generated.

Let us assume that the target variables and the inputs are related via the equation:

$$y^{(i)} = \theta^T x^{(i)} + \epsilon^{(i)} \tag{1}$$

Where:

- $y^{(i)}$ is the target variable for the i-th example.

- θ represents the weights (parameters).
- $x^{(i)}$ is the input feature vector.
- $\epsilon^{(i)}$ is the error term.

The error term $\epsilon^{(i)}$ captures either unmodeled effects (such as features pertinent to predicting housing prices that we left out) or random noise.

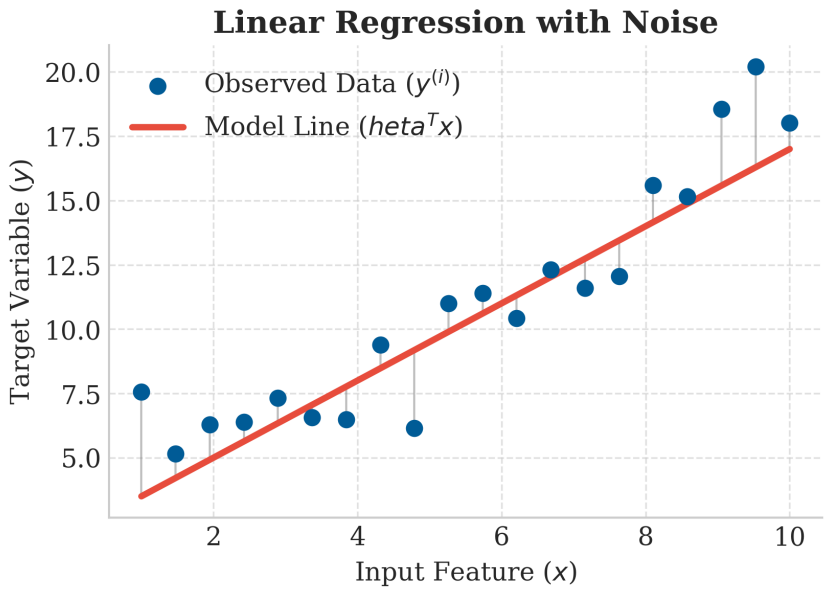

Figure 1: Visualizing the Linear Model: Data points deviate from the line due to noise $\epsilon^{(i)}$.

The Gaussian Assumption

The noise is assumed to be Gaussian, independent, and identically distributed (IID):

$$\epsilon^{(i)} \sim \mathcal{N}(0, \sigma^2)$$

In our regression model, we assume that the "noise" or error term $\epsilon^{(i)}$ for each data point is distributed according to a Gaussian distribution with a mean of 0 and variance σ^2. The probability density function is:

$$p(\epsilon^{(i)}) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(\epsilon^{(i)})^2}{2\sigma^2}\right) \tag{2}$$

This single assumption determines everything that follows.

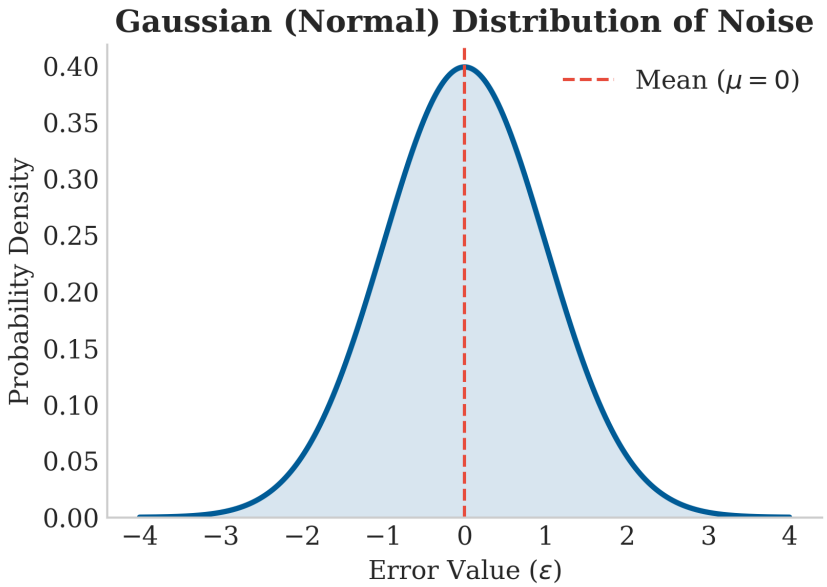

Figure 2: The Gaussian (Normal) Distribution of the Error Term.

3 From Gaussian Noise to Likelihood

If the noise is Gaussian, then the conditional distribution of $y^{(i)}$ given $x^{(i)}$ is determined by substituting $\epsilon^{(i)} = y^{(i)} - \theta^T x^{(i)}$ into the Gaussian density:

$$p(y^{(i)} \mid x^{(i)}; \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2}\right) \tag{3}$$

Focus on the term inside the exponent: $(y^{(i)} - \theta^T x^{(i)})^2$. This is precisely the **squared error**. Since it appears with a negative sign in the exponent, maximizing the probability is equivalent to minimizing this squared error.

To obtain the likelihood of the entire dataset, we multiply the individual probabilities for each data point from $i = 1$ to m. This product gives us the joint likelihood of observing all the data under the model:

$$L(\theta) = \prod_{i=1}^{m} p(y^{(i)} \mid x^{(i)}; \theta) \tag{4}$$

Substituting the Gaussian density:

$$L(\theta) = \prod_{i=1}^{m} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2}\right) \tag{5}$$

Our main objective is to maximize this likelihood function $L(\theta)$ to find the parameters θ that make the observed data most probable.

The Log-Likelihood

To find the maximum of a function, we typically take its derivative and set it to zero. However, taking the derivative of a product of m terms is computationally difficult. The derivation becomes much simpler if we maximize the **log-likelihood**, $\ell(\theta) = \log L(\theta)$, instead.

$$\begin{aligned}
\ell(\theta) &= \log \left[\prod_{i=1}^{m} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2} \right) \right] \\
&= \sum_{i=1}^{m} \log \left(\frac{1}{\sqrt{2\pi}\sigma} \right) + \sum_{i=1}^{m} \log \left(\exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2} \right) \right) \\
&= m \log \left(\frac{1}{\sqrt{2\pi}\sigma} \right) - \frac{1}{2\sigma^2} \sum_{i=1}^{m} (y^{(i)} - \theta^T x^{(i)})^2
\end{aligned}$$

Notice the structure of this final equation:

1. The first term $m \log(\dots)$ is a constant with respect to θ.
2. To maximize $\ell(\theta)$, we must maximize the negative term on the right.
3. Maximizing a negative quantity is equivalent to **minimizing the positive quantity**:

$$\sum_{i=1}^{m} (y^{(i)} - \theta^T x^{(i)})^2$$

This is exactly the **Least Squares Cost Function** $J(\theta)$.

4 The Real Answer: Why Squared Error?

We now see the answer clearly. Linear regression minimizes squared error because:

> **Under the assumption of Gaussian noise, the Maximum Likelihood Estimator (MLE) is exactly the Least Squares method.**

It is not an arbitrary design choice. It is a statistical consequence. The square appears because the Gaussian density contains a quadratic term in the exponent. If the noise distribution were different (e.g., Laplacian), the appropriate loss function would change (e.g., Absolute Error).

5 Final Insight

Linear regression is usually introduced as:

"Fit a line by minimizing squared error."

But the deeper truth is:

- It assumes a generative model.
- It assumes Gaussian noise.
- It performs Maximum Likelihood Estimation.
- It produces least squares naturally.

The cost function is not the starting point. The probability model is. And that is why squared error appears not as a design choice, but as a derivation.

Note from the Author

This article is heavily influenced by lecture notes from Stanford and Princeton, along with my own mathematical understanding of the subject. My goal was to present the ideas in a simpler and more intuitive manner while preserving the rigor of the original derivations. The explanations and equations have been carefully reviewed to ensure correctness.